UNIVERSIDADE DE SÃO PAULO
INSTITUTO DE FÍSICA DE SÃO CARLOS

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Thermal Relaxation Asymmetry in Quantum Systems

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Thermal Relaxation Asymmetry in Quantum Systems

Dissertation presented to the Graduate Program in Physics at the Instituto de Física de São Carlos da Universidade de São Paulo, to obtain the degree of Master in Science.

Concentration area: Theoretical and Experimental Physics

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ACKNOWLEDGEMENTS

Primeira frase do agradecimento ....

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Última frase ....

"O estudo, a busca da verdade e da beleza são domínios em que nos é consentido sermos crianças por toda a vida."

Albert Einstein

ABSTRACT

RAUPP, C. **Thermal Relaxation Asymmetry in Quantum Systems**. 2026. 59 p. Dissertation (Master in Science) - Instituto de Física de São Carlos, Universidade de São Paulo, São Carlos, 2026.

This is the english abstract.

Keywords: LaTeX. USPSC class. Thesis. Dissertation. Conclusion course paper. Report.

RESUMO

RAUPP, C. **Thermal Relaxation Asymmetry in Quantum Systems**. 2026. 59 p. Dissertação (Mestrado em Ciências) - Instituto de Física de São Carlos, Universidade de São Paulo, São Carlos, 2026.

O resumo deve ressaltar o objetivo, o método, os resultados e as conclusões do documento. A ordem e a extensão destes itens dependem do tipo de resumo (informativo ou indicativo) e do tratamento que cada item recebe no documento original. O resumo deve ser precedido da referência do documento, com exceção do resumo inserido no próprio documento. (. . .) Salientamos que algumas Unidades exigem o titulo dos trabalhos acadêmicos em inglês, tornando necessário a inclusão das referências nos resumos e abstracts, o que foi adotado no **Modelo para TCC em LaTeX utilizando a classe USPSC** e no **Modelo para teses e dissertações em LaTeX utilizando a classe USPSC**. As palavras-chave devem figurar logo abaixo do resumo, antecedidas da expressão Palavras-chave:, separadas entre si por ponto e finalizadas também por ponto (**?**).

Palavras-chave: LaTeX. Classe USPSC. Tese. Dissertação. Trabalho de conclusão de curso (TCC). Relatório.

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LIST OF FRAMES

LIST OF ABBREVIATIONS AND ACRONYMS

ABNT	Associação Brasileira de Normas Técnicas
abnTeX	ABsurdas Normas para TeX
IBGE	Instituto Brasileiro de Geografia e Estatística
LaTeX	Lamport TeX
USP	Universidade de São Paulo
USPSC	Campus USP de São Carlos

LIST OF SYMBOLS

Γ	Letra grega Gama
Λ	Lambda
ζ	Letra grega minúscula zeta
$\in$	Pertence

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3 QUANTUM THERMODYNAMICS

3.1 Quantum Correlations

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3.4 Relaxation, Entropy Production and Irreversibility

3.4.1 Thermal Kinematics

4 FROM DISCRETE TO CONTINUOUS QUANTUM SYSTEMS

Nature has a diversity of physical phenomena, being some better described by discrete models while others are better described by coninuous models. Quantum mechanics was formulated in both discrete and continuous regimes as the Heisenberg and Schrödinger frameworks, which, despite of having fundamentaly different mathematical resources, complements each other when explaining experimental results. Finding the same phenomena in different regimes not only brings to light fundamental aspects of nature but also increases the possiblity of technological applications. For this reason, this chapter explores the thermal relaxation asymmetry for discrete and continuous-variable quantum systems.

Given that this work aims to understand how quantum resources affect the heating and cooling asymemtry, we examine specific models that have these resources. Among a variety of possible discrete models, we chose the two interacting qubits in contact with the same bosonic bath. This model allow us to analyse the effects of quantum correlations and non-Markovianity in the thermal relaxation. While for continuous-variables models, the natural choice was Gaussian states for its simplicity and experimental accessibility. For this case, the quantum resources we explore are displacement and squeezing of thermal states.

4.1 Two Interacting Qubits

The interaction between N qubits with the same energy difference (h) can be described by a family of Hamiltoninans

$$H_S = -h\sum_{j=1}^{N}\sigma_j^z - J\sum_{j=1}^{N-1}[(1+\gamma)\sigma_j^x\sigma_{j+1}^x + (1-\gamma)\sigma_j^y\sigma_{j+1}^y + \Delta\sigma_j^z\sigma_{j+1}^z] \tag{4.1}$$

where σ_k^x, σ_k^y, and σ_k^z are the Pauli matrices for the k-th qubit. J is the coupling parameter between the qubits, while γ and Δ creates anisotropries to the system. Depending on the values of the anisotropic paramters, we can distinguish different models (1) as shown in Table 1.

Table 1 – Model's Parameters

Parameters	Models
$\Delta = 1,\ \gamma = 0$	XXX
$\Delta \neq 0,\ \gamma = 0$	XXZ
$\Delta \neq 0,\ -1 < \gamma < 1$	XYZ
$\Delta = 0,\ \gamma = 0$	XX
$\Delta = 0,\ -1 < \gamma < 1$	XY
$\Delta = 0,\ \gamma = \pm 1$	Transverse Field Ising (TFI)

What matters for us about this models is that the different interaction cases will lead to different correlations between the qubits. These correlations will be measured by the mutual information and entanglement of formation, defined in Section 3.1. Note that not all of these models can create entanglement, which we will discuss later in this section.

4.1.1 Thermal Relaxation of Two Interacting Qubits

Lets start considering two qubits coupled to the same bosonic bath, as shown in Figure 1. In this first analysis, the qubits has an energy gap of $\frac{\omega_0}{2}$ and exchange energy according to the model XX (with $\Delta = 0$ and $\gamma = 0$), where the interaction strength is given by J. One qubit will heat up from the temperature T_c while the other will cool down from the temperature T_h, being the bath at an intermediate temperature T_w ($T_c < T_w < T_h$).

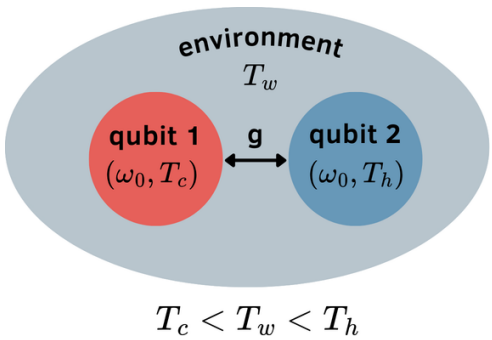

Figure 1 – Model for two interacting qubits with a global bath.

Depending on the initial state of the system, the relaxation process will differ. In our case, the only variable to characterize the qubit's initial state is the temperature. We could add coherences to the qubits, but this would make the terms 'heating' (or 'cooling') inaccurate due to the inexistence of thermal states. So, we will keep the qubits locally thermal while adding global correlations if desired.

To make a fair comparison between heating and cooling, we need to find statistically equivalent initial states with different temperatures. Because the asymemtry appears only for relaxation of nonequilibrium states, it is natural to distinguish them according to how far from equilibrium it is. Thereby, the relative entropy, which is related to the

nonequilibrium free energy, seems the most approprieate choice of distinguishability measure. That said, statistically equivalent initial states for each qubit must have the same initial relative entropy from the local steady state of the dynamics (Equation 4.2). However, to find the appropriate initial temperature, first, we must find the local steady state (ρ_{ss}^{qubit}).

$$K[\rho_c||\rho_{ss}^{qubit}] = K[\rho_h||\rho_{ss}^{qubit}] \tag{4.2}$$

According to its definition, a steady state do not change in time, which mathematically can be found by solving $\mathcal{L}(\rho_{ss}) = 0$, where ρ_{ss} is the global steady state. Despite this being the most traditional approach, there is another simpler way. First, we know that the steady state of a Davies Map is always a Gibbs state, which allow us to assume that

$$\rho_{ss} = \frac{e^{-\beta_w H_S}}{\mathcal{Z}}, \tag{4.3}$$

where β_w is the inverse temperature of the bath and H_S is the total system hamiltonian 4.1. Then, we can trace out one of the qubits* and write the expression in terms of H_S eigenvalues and eigenvectors to obtain

$$\rho_{qubit} = \begin{bmatrix} e^{\beta_w \omega_0} + \cosh(\beta_w J) & 0 \\ 0 & e^{-\beta_w \omega_0} + \cosh(\beta_w J) \end{bmatrix}. \tag{4.4}$$

Because every two-level system in a Gibbs state has the same format, we can generally isolate the qubit's temperature, resulting in

$$\beta_{qubit}^{f} = \frac{1}{\omega_0} \ln \left(\frac{e^{\beta_w \omega_0} + \cosh(\beta_w J)}{e^{-\beta_w \omega_0} + \cosh(\beta_w J)} \right). \tag{4.5}$$

Note that, even though the global system thermalizes to the bath's temperature, the qubits' local thermal state correspond to a different temperature. This happens because the interaction between the qubtis generates correlations that are erased by the partial trace. Now, we can solve Equation 4.2 and find the initial temperatures.

Because we want to understand how quantum correlations may affect the thermal relaxation asymmetry, the next step is to find out what is the maximum correlation possible for our initial state. Lets assume the initial global state has the form of an X-state,

* It does not matter which one, because both equilibrate to the same temperature

$$\rho = \rho_c \otimes \rho_h + \chi = \begin{bmatrix} a & 0 & 0 & f \\ 0 & b & e & 0 \\ 0 & e^* & c & 0 \\ f^* & 0 & 0 & d \end{bmatrix}, \tag{4.6}$$

where the diagonal elements are

$$a = \frac{e^{\omega_0(\beta_c+\beta_h)/2}}{\mathcal{Z}_c\mathcal{Z}_h} \tag{4.7}$$

$$b = \frac{e^{\omega_0(\beta_c-\beta_h)/2}}{\mathcal{Z}_c\mathcal{Z}_h} \tag{4.8}$$

$$c = \frac{e^{-\omega_0(\beta_c-\beta_h)/2}}{\mathcal{Z}_c\mathcal{Z}_h} \tag{4.9}$$

$$d = \frac{e^{-\omega_0(\beta_c+\beta_h)/2}}{\mathcal{Z}_c\mathcal{Z}_h} \tag{4.10}$$

where $\mathcal{Z}_c = 2\cosh(\omega_0\beta_c/2)$, $\mathcal{Z}_h = 2\cosh(\omega_0\beta_h/2)$ and the off-diagonal elements are

$$\chi = e\,|01\rangle\langle 10| + e^*\,|10\rangle\langle 01| + f\,|00\rangle\langle 11| + f^*\,|11\rangle\langle 00|\,. \tag{4.11}$$

For ρ to be considered a proper density matrix it must satisfy all the properties: (1) hermiticity, (2) unit trace and (3) positive semi-definite. So, we must choose e and f values that preserve property (3) of ρ, which means that all eigenvalues of ρ must be non-negative ($\lambda_i \geq 0$ for all i). When diagolizing ρ, we get two identical quadratic equations of the form

$$\lambda = \frac{x + y \pm \sqrt{x^2 + y^2 - 2xy + 4z}}{2} \geq 0 \tag{4.12}$$

where the variables can be $x = (b, a)$, $y = (c, d)$, $z = (|e|^2, |f|^2)$. Because the postive sign will always satisfy the inequality, we will analyse the restriction this inequality brings when choosing the negative sign. Manipulating the expression, we find $z \leq xy$ that means

$$|e| \leq \frac{1}{\mathcal{Z}_c\mathcal{Z}_h} \tag{4.13}$$

$$|f| \leq \frac{1}{\mathcal{Z}_c\mathcal{Z}_h}. \tag{4.14}$$

Now that we know the initial temperatures and maximum initial correlations, we can solve the Lindblad master equation to obtain the density matrices for the global

system and each individual qubit as a function of time. After computing the Quantum Fisher Information, we solve the thermal kinematic's variables as described in Section 3.4.1.

The comparison between the degree of completion of the heating qubit (red) and the cooling qubit (blue) is shown in Figure 2, where the dashed lines represent the case with no interaction between the qubits and the continuous lines represent the interacting case. Note that in our results, heating is always faster than cooling, because the red degree of completion is above the correspondent blue lines. Another result we can extract is that, when qubits can interact between themselves, they relaxate faster. Furthermore, the presence of initial correlations seems to delay the dynamics when we compare the continuous lines of both graphs.

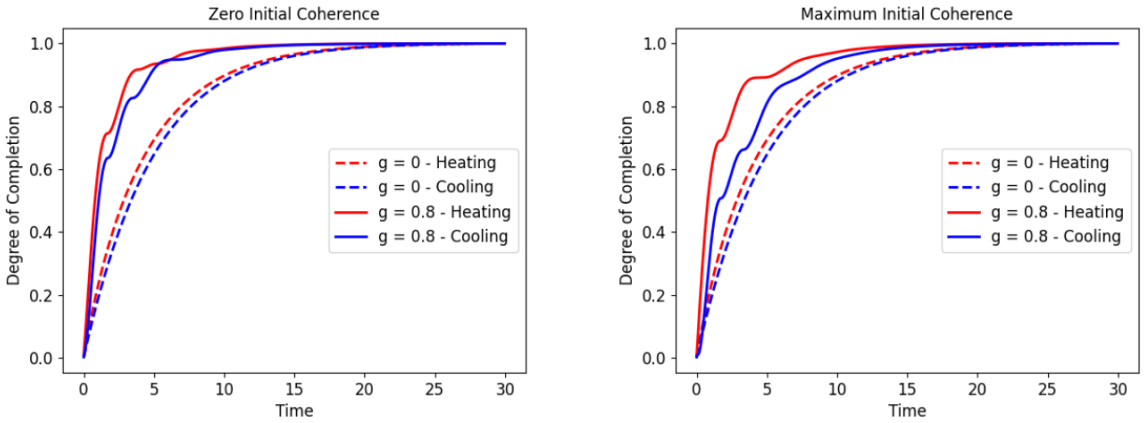

Figure 2 – Degree of completion for XX-model.

4.2 Gaussian States

4.2.1 Thermal Relaxation of Gaussian States

4.2.1.1 Asymmetry for Thermal States

4.2.1.2 Asymmetry for Displaced States

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5 CONCLUSÃO

REFERENCES

1 MEDINA, I. *et al.* Variational-quantum-eigensolver–inspired optimization for spin-chain work extraction. **Phys. Rev. A**, American Physical Society, v. 110, p. 012443, Jul 2024. Available at: https://link.aps.org/doi/10.1103/PhysRevA.110.012443.

APPENDIX

APPENDIX A – APÊNDICE(S)

Elemento opcional, que consiste em texto ou documento elaborado pelo autor, a fim de complementar sua argumentação, conforme a ABNT NBR 14724 (**?**).

Os apêndices devem ser identificados por letras maiúsculas consecutivas, seguidas de hífen e pelos respectivos títulos. Excepcionalmente, utilizam-se letras maiúsculas dobradas na identificação dos apêndices, quando esgotadas as 26 letras do alfabeto. A paginação deve ser contínua, dando seguimento ao texto principal. (**?**)

APPENDIX B – EXEMPLO DE TABELA CENTRALIZADA VERTICALMENTE E HORIZONTALMENTE

A Tabela 2 exemplifica como proceder para obter uma tabela centralizada verticalmente e horizontalmente.

Table 2 – Exemplo de tabela centralizada verticalmente e horizontalmente

Coluna A	**Coluna B**
Coluna A, Linha 1	Este é um texto bem maior para exemplificar como é centralizado verticalmente e horizontalmente na tabela. Segundo parágrafo para verificar como fica na tabela
Quando o texto da coluna A, linha 2 é bem maior do que o das demais colunas	Coluna B, linha 2

Fonte: Elaborada pelos autores.

APPENDIX C – EXEMPLO DE TABELA COM GRADE

A Tabela 3 exemplifica a inclusão de traços estruturadores de conteúdo para melhor compreensão do conteúdo da tabela, em conformidade com as normas de apresentação tabular do IBGE.

Table 3 – Exemplo de tabelas com grade

Coluna A	**Coluna B**
A1	B1
A2	B2
A3	B3
A4	B4

Fonte: Elaborada pelos autores.

ANNEX

ANNEX A – EXEMPLO DE ANEXO

Elemento opcional, que consiste em um texto ou documento não elaborado pelo autor, que serve de fundamentação, comprovação e ilustração, conforme a ABNT NBR 14724. (**?**).

O **ANEXO B** exemplifica como incluir um anexo em pdf.

ANNEX B – ACENTUAÇÃO (MODO TEXTO - LaTeX)

Figure 3 – Acentuação (modo texto - LaTeX)

\'a - á
\'a - à
\~a - ã
\^a - â
\'e - é
\^e - ê
\'{\i} - í
\'I - Í
\'o - ó
\~o - õ
\^o - ô
\'u - ú
\"u - ü
\c{c} - ç
\c{C} - Ç

Fonte: Comandos [...] (**??**)

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tabelas, 49, 51